

AI Powered Academic Research Assistant

Divyesh Mistry
St. Francis Institute of Technology
Computer Engineering
Mumbai, India
divyeshrmistry@gmail.com

Prem Tatkar
St. Francis Institute of Technology
Computer Engineering
Mumbai, India
tatkariprem2002@gmail.com

Daniel Ferreira
St. Francis Institute of Technology
Computer Engineering
Mumbai, India
danferreira376@gmail.com

Kyran Almeida
St. Francis Institute of Technology
Computer Engineering
Mumbai, India
almeidakyran@gmail.com

Abstract—In the expanding world of research, there is a growing need, for intelligent and efficient tools that can help researchers navigate through vast amounts of information. This project introduces an AI-powered Academic Research Assistant to revolutionize the research process. By utilizing cutting-edge natural language processing and machine learning techniques this assistant seamlessly integrates into a researcher’s workflow. It can potentially transform the landscape of research by providing a solution, to information overload. This assistant enhances research efficiency promotes knowledge sharing and encourages collaboration across disciplines. Not only does this project showcase technological innovation. It also highlights how AI can complement human intelligence in tackling complex knowledge-based challenges.

Index Terms—AI, Academic Research Assistant, Natural Language Processing

I. INTRODUCTION

An Academic Research Assistant is a crucial role in the field of academic research, providing support to faculty members and research teams. They assist in conducting literature reviews, gathering and analyzing data, and preparing reports or manuscripts for publication. Research assistants often work closely with professors or researchers, helping them design experiments, develop research methodologies, and formulate research questions. Our project aims to provide researchers with an AI-powered academic research assistant. It will ease the process of information gathering for researchers.

II. RELATED WORK

A. LongT5

LongT5 is a new text-to-text transformer specially designed to handle longer sequences. This transformer is based on the T5 architecture but with several modifications to improve its performance with longer sequences. These changes include a more efficient attention mechanism for extended sequences, pre-training LongT5 with a dataset of lengthy sequences, and increasing the model’s size. Evaluations of LongT5 on various summarization and question-answering tasks show that it outperforms the original T5 model and achieves state-of-the-art results for some tasks. However, its usage is not as

widespread as other language models, resulting in limited research on its advantages and drawbacks.[1]

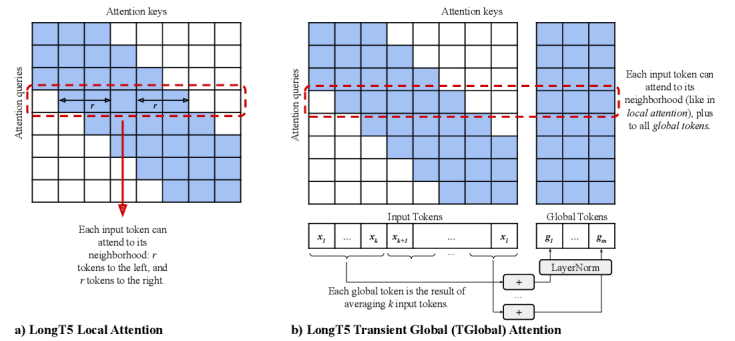


Fig. 1. Illustration of the two attention mechanisms experimented with in LongT5.[1]

B. PEGASUS

PEGASUS introduces a new pre-training method for abstractive summarization. Building upon the Transformer language model, PEGASUS features several modifications to improve abstractive summarization performance. These enhancements include utilizing a new objective function promoting informative and fluent summaries and pre-training PEGASUS on a collection of human-authored summaries along with their corresponding source documents. Evaluations show PEGASUS surpassing state-of-the-art results on multiple summarization tasks.[2]

III. PROPOSED WORK

The proposed work is to design an AI-powered academic research assistant that utilizes text-scraping, machine learning, and natural language processing techniques to enhance the research process. The system will scrape relevant data from academic websites, and journals ensuring comprehensive coverage of research sources. The project will culminate in the development of a user-friendly web interface where users can input queries, browse search results, and explore papers.

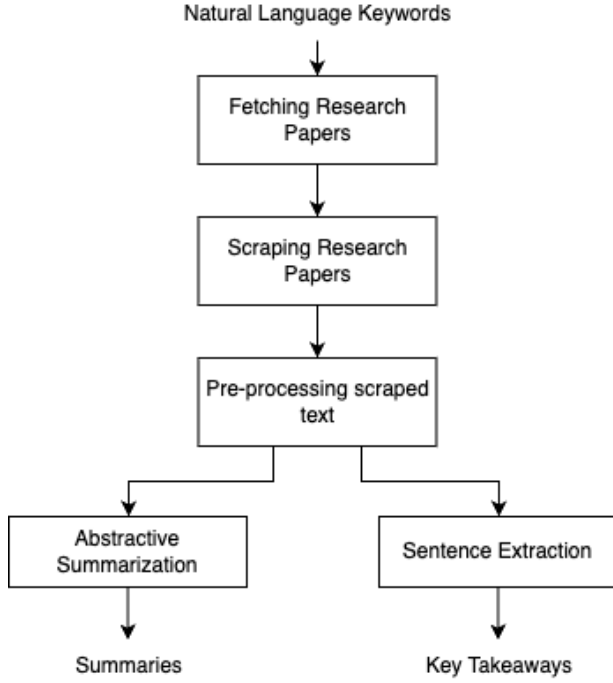


Fig. 2. Block diagram of AI-Powered Academic Research Assistant.

A. Fetching Research Papers

Utilizing the arXiv API, we facilitate the fetching of research papers based on specified criteria. The arXiv API allows users to search for papers using keywords, authors, categories, or specific identifiers. The process involves constructing a query string with parameters such as search terms, date range, or categories, and sending it to the arXiv server. The server then responds with a list of papers that match the criteria. These research papers are downloaded in a PDF format and further scraped. Thus by utilizing the arXiv API, researchers can efficiently retrieve relevant research papers for further analysis or review. This method simplifies accessing scientific literature, keeping researchers up-to-date with advancements in their field. Moreover, integrating the arXiv API into applications or tools enables automation and streamlines the retrieval process, saving time and effort for the researchers.

B. Scraping Research Papers

By utilizing the ‘fitz’ library in Python, we facilitate text scraping from the previously downloaded PDFs. The ‘fitz’ library provides us with the necessary functionalities to work with PDF files, including extracting text, metadata, and annotations. The process involves loading the PDF document using ‘fitz’ iterating through its pages then extracting text from each page. Once the text is extracted, it can be analyzed further for summarization. Thus by utilizing the ‘fitz’ library, we automate the process of extracting text from research paper PDFs, enabling efficient analysis and information retrieval.

C. Pre-processing Scraped Text

The scraped data might contain formulas or image tags that are not displayed in proper format. Therefore there is a need to clean the text Data and get it correct format. This is done by passing the scraped text into the pre-processing function, where with the help of multiple Regular Expressions the text is cleaned. The use of Regular Expressions (regex) allows us to manipulate the data according to our needs and thus allows us to remove any redundant form of text that may reduce the accuracy of the final summary generated by the transformer.

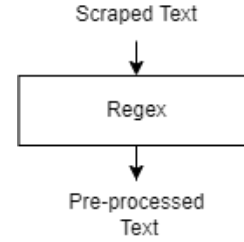


Fig. 3. Preprocessing the Scraped Data

D. Sentence Extraction

Find out key takeaways from all the papers. These key takeaways give us an idea about the essential sentences from the papers. We use the LexRank algorithm for this very purpose. LexRank processes the text to identify important sentences based on their semantic similarity and centrality within the document. The algorithm calculates the importance of each sentence using the formula:

$$S_i = (1 - \lambda) + \lambda \sum_{j=1}^n \frac{w_{ji}}{d_j} S_j \quad (1)$$

Where:

- S_i is the importance score of sentence i ,
- λ is a damping factor (typically set to 0.85),
- w_{ji} represents the weight of the edge between sentences i and j in the similarity graph,
- d_j is the out-degree of sentence j , i.e., the number of sentences connected to sentence j ,
- n is the total number of sentences in the document.

E. Abstractive Summary

1) *Extractive Summarization*: Extractive methods attempt to summarize articles by identifying the important sentences or phrases from the original text and stitch together portions of the content to produce a condensed version. These extracted sentences are then used to form the summary.[9]

2) *Abstractive Summarization*: This technique, unlike extraction, relies on being able to paraphrase and shorten parts of a document using advanced natural language techniques. Abstractive machine learning algorithms can generate new phrases and sentences to capture the meaning of the source document. When such abstraction is done correctly in deep

learning problems, they can assist in overcoming grammatical inaccuracies.[9]

On thorough evaluation of both the summary models, we use the abstraction model. The summary is generated using the LongT5 Transformer model. The LongT5 model works by first encoding the input into a sequence of tokens. These tokens are then passed through the transformer encoder, which learns to identify the relationship between the tokens. The encoded tokens are then passed through the transformer decoder which generates the output summary. This summary includes important information from the paper in a concise manner.

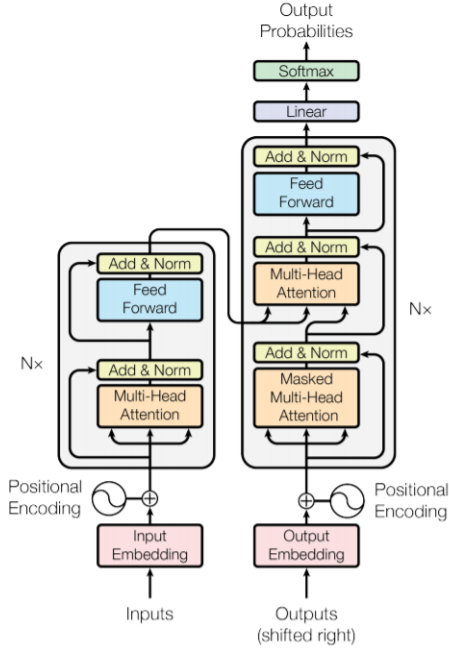


Fig. 4. Transformer architecture.[4]

Fig. 4 shows the architecture of a transformer. LongT5's architecture is similar to this one. Encoders in Transformers, also known as the transformer encoder, are a key component of the transformer encoder-decoder architecture. They are responsible for analyzing and representing the input sequence in a way the model can understand. The encoder processes the input sequence and produces a continuous representation, or embedding, of the input. These embeddings are then passed to the decoder to generate the output sequence. Each layer of the encoder contains a self-attention mechanism that allows the model to weigh the importance of different input sequence parts by calculating the embeddings' dot product. This mechanism is also known as multi-head attention.

The decoder also typically consists of multiple layers, including a self-attention mechanism and a feed-forward network. The decoder uses the embeddings produced by the encoder and its internal states to generate the output sequence.

Global-local attention is a type of attention mechanism used in the ETC architecture. ETC receives two separate input sequences: the global input and the long input. Typically, the

long input contains the input a standard Transformer would receive, while the global input contains a much smaller number of auxiliary tokens.[7]

LongT5 uses an attention mechanism called Transient Global(TGlobal), which mimics ETC's Global-Local Attention mechanism, but without requiring additional side inputs.

IV. RESULTS AND DISCUSSION

A. Results

The Research Assistant starts by letting the user input their desired topic as seen in Fig. 5, or the user can upload a PDF from their local system to Summarize and find its Key takeaways.

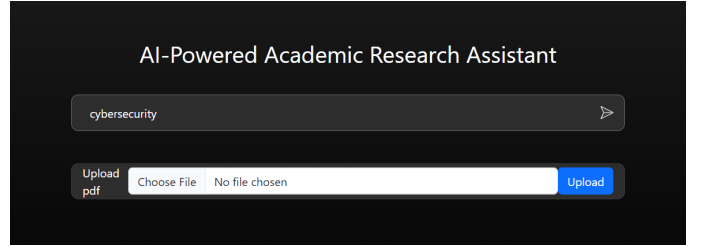


Fig. 5. GUI Page of AI-Powered Academic Research Assistant.

After receiving the input query from the user, the system will pass the input to the arXiv API to retrieve and download the papers in PDF format. The arXiv API has been modified to include papers only from the "Computer Science" field to avoid any ambiguities related to the topics.

"As discussed in section III. B", the text is scraped from the downloaded PDFs and then fed into the LongT5 transformer model which gives the summaries for each paper, refer Fig. 6.



Fig. 6. Summary Page of AI-Powered Academic Research Assistant.

Implementation of the LexRank algorithm (Refer Section III.D) enables the system to compare and rank the data present and produce the key takeaways as shown in Fig. 7. A damping factor set to the default value of 0.85, helps the model to balance the influence of direct connections (edges) between nodes and the probability of transitioning to any node in the graph randomly.

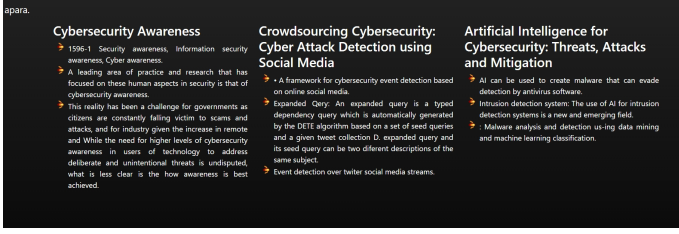


Fig. 7. Key Takeaway Page of AI-Powered Academic Research Assistant.

B. Analysis

To assess the performance and effectiveness of the model, we use common evaluation metrics of ROUGE-1, ROUGE-2, and ROUGE-L. ROUGE, or Recall-Oriented Understudy for Gisting Evaluation, is a set of metrics and a software package used for evaluating automatic summarization and machine translation software in natural language processing. The metrics compare an automatically produced summary or translation against a reference or a set of references (human-produced) summary or translation.[8]

TABLE I
TRAINING AND VALIDATION LOSS

Epoch	Training Loss	Validation Loss
1	1.0771	1.059394
2	1.0193	1.053011
3	0.9545	1.049587

TABLE I shows the model gradually reducing the difference between its predictions and the known correct outputs for the training data, which is reflected in the decreasing training loss. The validation loss also decreases over time, which indicates that the model is learning effectively.

TABLE II
ROUGE SCORE

Rouge1	Rouge2	RougeL	RougeLsum	Gen Len
39.2592	13.6097	25.1508	35.285	111.22
39.1521	14.8550	25.4706	35.3713	110.14
39.8436	14.7630	25.6676	36.4820	103.02

TABLE II shows the different ROUGE scores that measure different aspects of how similar the generated summary is to the reference summaries. The Gen Len is the average length of the generated summaries. The scores in the table suggest that the model's performance improved as the number of epochs increased.

V. CONCLUSION

The AI-Powered Academic Research Assistant with document scraping abilities has the potential to be a critical tool for researchers, significantly reducing the effort and time spent on information retrieval, data analysis, and literature review. This research assistant aims to empower academics and contribute

to improvements in numerous fields of study by leveraging document scraping and AI/ML techniques.

REFERENCES

- [1] Guo, Mandy & Ainslie, Joshua & Uthus, David & Ontanon, Santiago & Ni, Jianmo & Sung, Yun-Hsuan & Yang, Yinfei. "LongT5: Efficient Text-To-Text Transformer for Long Sequences." arXiv:2112.07916 [cs.CL] 2021.
- [2] Zhang, Jingqing & Zhao, Yao & Saleh, Mohammad & Liu, Peter. "PEGASUS: Pre-training with Extracted Gap-sentences for Abstractive Summarization." IEEE 2020
- [3] Subramanian, Sandeep & Li, Raymond & Pilault, Jonathan & Pal, Christopher. "On Extractive and Abstractive Neural Document Summarization with Transformer Language Models." IEEE 2020.
- [4] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. "Attention is all you need." In Advances in Neural Information Processing Systems, Volume 30 arXiv:1706.03762v7 [cs.CL] 2023
- [5] Mihalcea, Rada & Tarau, Paul. "TextRank: Bringing Order into Texts" (2004)
- [6] Joshua Ainslie, Santiago Ontanon, Chris Alberti, Vaclav Cvicek, Zachary Fisher, Philip Pham, Anirudh Ravula, Sumit Sanghai, Qifan Wang, Li Yang. "ETC: Encoding Long and Structured Inputs in Transformers." arXiv:2004.08483v5 [cs.LG] 2020
- [7] <https://avinetworks.com/glossary/web-scraping/> 2023/10/16
- [8] [https://en.wikipedia.org/wiki/ROUGE_\(metric\)](https://en.wikipedia.org/wiki/ROUGE_(metric)) 2023/10/16
- [9] <https://www.impelsys.com/blog/an-overview-of-text-summarization-in-natural-language-processing/> 2023/10/16
- [10] Gunes Erkan & Dragomir R. Radev "LexRank: Graph-based Lexical Centrality as Salience in Text Summarization", arXiv:1109.2128v2 [cs.CL] 2011